DS701 Final Team Project

Assigned Wed Nov 11 · due Wed Dec 2, 11:59 pm · checkpoint Thur & Mon Nov 19 & 23 and final orals Thurs - Fri Dec 3-4

Teams of three (two by exception). Your final team may differ from your midterm team. If you cannot find teammates, fill out the team-matching form posted on Piazza by **Fri Nov 13** and course staff will help match you.

Grading is 70% team, 30% individual. The team 70% covers the deliverable and your team's overview at the final oral. The individual 30% comes from **two oral checkpoints**.

Step	Date	Form	Examines	Weight
Written check	Due by your oral checkpoint 1 **Thurs or Mon Nov 19 or 23**	push to our GitHub repo	questions, causal diagram	feedback only
Checkpoint 1 orals	**Thurs or Mon Nov 19 or 23**	15 or 20-min oral	causal design, first pass	10%
Final orals	**Thurs or Fri Dec 3 or 4**	15 or 20-min oral	results, what-ifs	20%

At each oral, start with a short 5-minute summary, then staff direct questions to each member individually — on Nov 19 or 23, and on Dec 3 or 4 (at least one about a part you worked on, one about how your approach would change under new conditions). Only the named member may answer. Questions are drawn live, so answers cannot be scripted. Each checkpoint score stands on its own. A member who cannot explain the team's own work receives a low individual score regardless of how good the deliverable is: the orals exist to reward understanding, not polish.

The task

Pick **one** dataset from the menu below. Pose **2–3 concrete questions** about it, answer them with techniques from this course, and **justify every modeling choice** — why this distance, this model family, this evaluation, and what you tried that didn't work. You now have the full toolbox: everything from the midterm (clustering, classification, regression, SVD/PCA) plus causal inference (L14), neural networks (L16), text and TF-IDF (L18), recommender systems (L19), and graphs (L20; centrality and community detection are in the optional Graphs II module). Each menu entry lists three starter questions of increasing ambition — you may use them, sharpen them, or replace them with your own of comparable depth.

One of your questions must be causal or counterfactual in flavor — a "would X have changed Y?" question, not just a prediction. Answer it with the tools of L14 *and with appropriate humility*: draw the causal diagram you are assuming, say what you adjusted for and why, and state explicitly **what you can and cannot claim** from observational data. A careful "we cannot conclude causation here, and this is precisely why" is a strong answer; a casual causal claim from a regression coefficient is a weak one. Expect oral questions on this.

AI policy. AI assistance is allowed and encouraged — this project is also practice in directing AI tools productively. But every AI-assisted artifact (code, analysis, prose) must be understood and justified by your team: the oral examination checks exactly this.

Deliverables

Everything is due **Wed Dec 2, 11:59 pm**, but see the timeline: we strongly advise being done by **Mon Nov 30**, because Dec 2 is also quiz-3 day.

1. **A private GitHub repository**, one per team. Create it as a **private** repository and invite the instructors' GitHub accounts `trgardos` and `Farid-Karimli` as collaborators so they can review it; do this when you create the repo, because staff review it from Nov 13 on. It contains:
   - a **runnable notebook**: executes top to bottom on a fresh kernel, random seed fixed, and a data-loading cell that fetches or reads the dataset — staff will run it;
   - `CONTRIBUTIONS.md`: a section per member stating specifically who did what. The oral draws on it — claim your work accurately.
2. **A report of at most 3 pages** (PDF, in the repo *and* submitted on Gradescope), organized per question: the question → your approach and why → the evidence → limitations. Include your causal-claims paragraph (what you can and cannot conclude). Figures count toward the 3 pages; references don't.
3. **The repo link**, submitted with the PDF on Gradescope.

Checkpoint pushes. Push to your repository by these deadlines:

- **Thurs or Mon Nov 19 or 23 before your oral checkpoint — written check.** `QUESTIONS.md` with your 2–3 questions and, for the causal one, the causal diagram you are assuming, the treatment, the outcome and what you plan to adjust for.

The final oral runs **Thurs or Fri Dec 3 or 4** by scheduled appointment.

Dataset menu — choose one

1. Yelp reviews — a text corpus (L18, plus classifiers and clustering)

10,000 Yelp reviews of 524 businesses by 5,679 users, 2005–2015: star rating (1–5), review text (median ~90 words), date, and `funny`/`useful`/`cool` vote counts. One JSON record per line, 9.1 MB. Download: https://tools4ds.github.io/ds701/data/review.json (load with `pd.read_json(..., lines=True)`).

Starter questions: (i) How well do TF-IDF features predict star rating (or low/mid/high), against a sensible baseline — and which terms carry the signal? (ii) Do the reviews cluster into interpretable topics, and how do topics relate to ratings and drift over the decade? (iii) *Causal:* longer reviews

get more "useful" votes — would writing a longer review *cause* more votes, or do user, business, and review-age confounds explain it? What can a 2005–2015 snapshot never tell you about that?

A strong project treats the text pipeline choices (tokenization, min-df, stop words, n-grams) as modeling decisions to defend, not defaults to accept.

2. MovieLens-100k — user–item ratings (L19, plus SVD and evaluation)

The classic recommender benchmark: 100,000 ratings (1–5) from 943 users on 1,682 movies, with user demographics and movie genres. 4.9 MB zip, stable for 25 years: `https://files.grouplens.org/datasets/movielens/ml-100k.zip` (`u.data` is tab-separated: user, item, rating, timestamp; see the bundled README).

Starter questions: (i) Compare rating predictors of increasing complexity — global/user/item-mean baselines, item–item collaborative filtering, matrix factorization — with an honest train/test split; where does complexity stop paying? (ii) Cold start: how does accuracy degrade with short user histories, and what would you actually show a brand-new user? (iii) *Causal:* ratings are not missing at random — users choose what to watch. Does that make popular movies look better than they are, and what *would* happen if your recommender started steering choices (feedback loops)? What can logged ratings alone support?

A strong project evaluates the way a recommender is used (per-user, ranking-aware), not just a global RMSE, and says which comparison the split design actually licenses.

3. Political blogs — a directed network (L20 + the optional Graphs II module)

Requires the optional module Graphs II: centrality and clustering (notes and slides; not on any quiz): the centrality and community-detection questions below use it. Choose this dataset only if your team is willing to work through that module on its own.

The 2004 U.S. political blogosphere (Adamic & Glance, 2005): 1,490 blogs, ~19,000 directed hyperlinks, each blog hand-labeled liberal or conservative (758/732). GML format, 951 KB. Download: `https://tools4ds.github.io/ds701/data/polblogs.gml` (load with `nx.read_gml(...)`; the node attribute `value` is the label — 0 liberal, 1 conservative).

Starter questions: (i) Characterize the structure: in/out-degree distributions, components, clustering — and do different centrality measures agree on who the hub blogs are? (ii) Run community detection *without* the labels: how well do discovered communities recover the political sides, and where do they disagree? (iii) *Counterfactual:* how much linking crosses ideological lines, and would removing the top bridge blogs have disconnected the two sides? Careful — a hyperlink is not influence; say what data you would need to claim influence.

A strong project respects directedness (a link *to* is not a link *from*) and quantifies label agreement properly instead of eyeballing a graph plot.

4. NSW job training — observational tabular data (L14)

The canonical causal-inference dataset (LaLonde 1986; Dehejia & Wahba 1999): did a 1970s job-training program raise 1978 earnings? You get the *randomized experiment* — 185 treated, 260 control (`nswre74_treated.txt`, `nswre74_control.txt`) — plus two *observational* comparison groups: `cps_controls.txt` (15,992) and `psid_controls.txt` (2,490). Whitespace- separated, no header; columns: treatment, age, education, black, hispanic, married, nodegree, earnings 1974,

1975, 1978. Under 2 MB total, from https://users.nber.org/~rdehejia/nswdata2.html (files at `https://users.nber.org/~rdehejia/data/<name>.txt`).

Starter questions: (i) What does the *experiment* say the training effect was, with uncertainty? (ii) Now discard the experimental controls and use CPS controls instead, as a real observational study would: what does the naive comparison say, and how close can adjustment — regression on covariates, or matching/stratifying on a logistic-regression propensity score — get to the experimental benchmark? (iii) *Counterfactual:* for whom would training have changed earnings? Say precisely which assumption (no unmeasured confounding) your subgroup claims lean on and where it is least plausible.

A strong project uses the experimental answer as the yardstick and is honest about which adjustments move toward it, which don't, and why.

How you are graded

Dimension	Scored	What a 4/4 looks like
Technique appropriateness	team	Methods fit the questions and the data; alternatives were considered and ruled out for stated reasons
Justification quality	team	Choices argued from the data and diagnostics — including every AI-assisted part
Result integrity	team	Claims match what the notebook shows; limitations (especially causal ones) stated plainly
Individual comprehension	per member	You can explain your part *and* the team's core method, and reason about a variation on the spot

Team score = mean of the three team dimensions (70%); your individual comprehension across the two oral checkpoints is your individual 30% (10% on checkpoint, 20% on final oral).

Suggested timeline (3.5 weeks)

- **Nov 11–19:** form the team (matching form by Fri Nov 13), set up the private repo and invite `trgardos` and `Farid-Karimli`, pick the dataset, run the loading cell, write your 2–3 questions *including the causal one* and draw its causal diagram. Push `QUESTIONS.md` by **Thurs or Fri Nov 19 or 23**.
- **Thurs or Mon Nov 19 or 23 — checkpoint 1** in section: causal design and first pass.
- **Nov 19–Dec 2** Complete your investigations and notebook development, iterate where the evidence is weak; draft the report; make the notebook run top-to-bottom on a fresh kernel.
- **Mon Dec 2:** Push the final notebook, report, and `CONTRIBUTIONS.md` to the repo and submit the repo link and PDF on Gradescope.

- **Thurs or Fri Dec 3 or 4**, the final oral. Every member should be able to explain every headline claim — rehearse by quizzing each other.